

An Adaptive Image-Processing Pipeline for Coronal-Hole Candidate Extraction from Full-Disk SDO/AIA 193 Å Images

Yi-Ting Hwang¹ · Chao-Wei Tsai¹ ·
Ya-Hui Yang²

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Abstract Coronal holes are low-emission regions in the solar corona and are important sources of high-speed solar-wind streams. Accurate and reproducible identification of coronal-hole structures from extreme-ultraviolet full-disk images is therefore essential for both solar-physics analysis and space-weather applications. However, automatic coronal-hole extraction remains challenging because of the high dynamic range of solar images, radial intensity variation across the solar disk, contamination from dark non-coronal-hole structures, and fragmented boundaries produced by threshold-based segmentation. This study proposes an adaptive coronal-hole candidate extraction pipeline for full-disk SDO/AIA 193 Å images. The proposed method integrates AIA intensity rescaling using an inverse hyperbolic sine transformation, solar-disk masking based on FITS header information, radial background intensity correction, adaptive K-Means thresholding, hybrid morphological filtering, and area-distance based connected-component selection. The method is designed to enhance low-intensity coronal-hole regions while reducing the influence of bright active regions, limb-related intensity variation, and small fragmented artifacts. Qualitative examples and comparisons with baseline segmentation methods show that the proposed pipeline provides more stable and structurally coherent coronal-hole candidate masks than fixed-threshold and standard automatic-threshold approaches. The resulting candidate masks provide a reproducible basis for visual assessment of coronal-hole-like low-emission structures and may support further structural or space-weather-related analyses in future work.

Keywords: Corona, Structures; Coronal Holes; Image Processing; Methods, Data Analysis; Solar Wind; Sun, UV Radiation

✉ Y.T. Hwang
hwangyt@gm.ntpu.edu.tw

¹ Department of Statistics, National Taipei University, New Taipei City, Taiwan

² National Central University, Taoyuan City, Taiwan

1. Introduction

Coronal holes are regions of reduced extreme-ultraviolet and X-ray emission in the solar corona. They are generally associated with open magnetic-field configurations and are recognized as important sources of high-speed solar-wind streams (Cranmer 2009; Rotter et al. 2012). Because the spatial distribution and temporal evolution of coronal holes are closely related to the geoeffective solar-wind environment, reproducible extraction of coronal-hole structures from full-disk solar images is an important task in solar physics and space-weather research.

Extreme-ultraviolet observations from the Atmospheric Imaging Assembly (AIA) onboard the Solar Dynamics Observatory (SDO) provide high-cadence full-disk solar images that are well suited for coronal-hole analysis (Lemen et al. 2012; Pesnell, Thompson, and Chamberlin 2012). Among the AIA channels, the 193 Å passband is frequently used for coronal-hole identification because coronal holes appear as relatively dark regions against the surrounding quiet-Sun and active-region background (Cranmer 2009; Krista and Gallagher 2009; Heinemann et al. 2019). Nevertheless, robust automatic segmentation remains nontrivial. The observed intensity distribution varies with solar activity, instrumental response, radial position on the solar disk, and the presence of bright active regions. In addition, other dark structures, such as filaments, may be confused with coronal holes when segmentation relies only on intensity information.

Traditional coronal-hole detection methods often depend on fixed intensity thresholds, cumulative intensity distributions, region-growing procedures, or empirical post-processing rules (Krista and Gallagher 2009; Rotter et al. 2012; Reiss et al. 2015). Although such methods are computationally efficient, their performance may deteriorate when the global intensity distribution changes across different observation times. More adaptive strategies, including clustering-based thresholding and machine-learning methods, have been introduced to reduce the dependence on manually specified thresholds (Reiss et al. 2015; Illarionov and Tlatov 2018; Heinemann et al. 2019). However, for practical applications, an effective coronal-hole extraction method should not only identify low-intensity regions, but also reduce disk-scale brightness variation, preserve structurally coherent regions, and remove small fragmented artifacts.

This study proposes an adaptive coronal-hole candidate extraction pipeline for full-disk SDO/AIA 193 Å images. The method combines intensity rescaling, dynamic solar-disk masking based on FITS header information, radial background correction, K-Means based adaptive thresholding, hybrid morphological filtering, and area-distance based connected-component selection. The main contribution of this study is not the introduction of a new clustering algorithm, but the development of a reproducible and physically motivated image-processing pipeline tailored to the extraction of coronal-hole-like low-emission structures in high-dynamic-range AIA images.

The objectives of this study are threefold. First, we construct a complete preprocessing and segmentation workflow for extracting coronal-hole candidate masks from SDO/AIA 193 Å observations. Second, we examine the contribution of each processing component through comparison with baseline segmentation methods and ablation analysis. Third, we demonstrate that the proposed

post-processing strategy can improve the structural coherence of extracted low-emission candidate regions while reducing fragmented artifacts and unstable segmentation behavior.

It should be noted that this study does not use manually annotated coronal-hole masks, magnetic-field data, solar-wind measurements, or external coronal-hole catalogs as inputs to the proposed segmentation method. Therefore, the detected regions should be interpreted as image-derived coronal-hole candidate regions rather than externally verified coronal-hole boundaries. The evaluation primarily emphasizes qualitative segmentation behavior, comparison with baseline image-processing methods, and ablation analysis of the processing components. In addition, supplementary reference-map comparisons with Solen.info indicated regions are included for representative calibrated cases to provide overlap-based agreement measures, including reference coverage, detected precision, intersection-over-union, and Dice coefficient.

The remainder of this article is organized as follows. Section 2 describes the observational data and preprocessing requirements. Section 3 presents the proposed coronal-hole candidate extraction method. Section 4 reports qualitative segmentation examples, baseline comparisons, supplementary reference-map evaluation, comparison with traditional morphological opening, and ablation analysis. Section 5 discusses the interpretation, limitations, and potential extensions of the method. Section 6 summarizes the main conclusions.

2. Data

This study focuses on the automatic extraction of coronal-hole candidate regions from full-disk solar extreme-ultraviolet images. Therefore, the data used in this study are limited to SDO/AIA 193 Å observations and their corresponding FITS header information. No solar-wind measurements, photospheric magnetic-field data, or external annotated coronal-hole catalogs are used as inputs to the proposed segmentation method. This design allows the study to examine the capability of a purely image-based pipeline to identify low-emission coronal-hole-like structures from AIA 193 Å intensity images.

2.1. SDO/AIA 193 Å Observations

The primary data source of this study is the Atmospheric Imaging Assembly (AIA) onboard the Solar Dynamics Observatory (SDO) (Lemen et al. 2012; Pesnell, Thompson, and Chamberlin 2012). AIA provides full-disk solar observations in multiple extreme-ultraviolet passbands. In this study, the 193 Å passband is selected because coronal holes generally appear as low-intensity regions at this wavelength, making it suitable for image-based coronal-hole candidate extraction.

The SDO/AIA 193 Å images were obtained from the Joint Science Operations Center (JSOC). The original image products are AIA Level 1 observations. Although Level 1 data have undergone basic instrumental calibration, they may still contain variations in solar-disk center position, image orientation, and

pixel scale. Therefore, before applying the proposed segmentation method, the images were further standardized to obtain consistent solar-disk geometry across observation times.

The observational period considered in this study spans from 2012 to 2024. Instead of using all available continuous SDO/AIA observations during this period, this study used a screened coronal-hole dataset constructed from coronal-hole cases identified within 2012–2024. After screening and quality control, the final dataset consisted of 94 screened coronal-hole cases. For each retained coronal-hole case, a complete temporal sequence of SDO/AIA 193 Å full-disk images covering the evolution of the corresponding coronal-hole structure was obtained and used as the input data for the proposed segmentation pipeline. Images were excluded if they contained severe observational artifacts, missing values, incomplete solar disks, abnormal FITS metadata, or insufficient image quality for representing the selected coronal-hole evolution.

For each image, the intensity matrix was used as the input for the proposed image-processing pipeline. The FITS header information was used to obtain the geometric properties of the solar disk, including the observed solar angular radius, pixel scale, and reference-pixel coordinates (Pence et al. 2010). These quantities are required for solar-disk mask construction and radial background correction.

2.2. Image Selection and Quality Control

The image dataset was constructed to include representative low-emission solar-disk structures under different observational conditions. Since this study aims to demonstrate the robustness of the proposed segmentation pipeline, the selected images include cases with large dark regions, fragmented low-emission boundaries, nearby active regions, and different positions on the solar disk.

Before segmentation, all images were checked for data completeness and geometric consistency. Images were excluded if the solar disk was incomplete, if the required FITS header keywords were unavailable, or if strong observational artifacts could affect the segmentation result. This quality-control procedure ensured that the analysis focused on the behavior of the proposed image-processing pipeline rather than on abnormal image-quality problems.

2.3. Data Usage in the Proposed Method

The proposed method uses only the SDO/AIA 193 Å image intensity and FITS header information. The image intensity is used for nonlinear intensity rescaling, radial background correction, adaptive thresholding, morphological filtering, and connected-component selection. The FITS header information is used to determine the solar-disk center, solar radius, and pixel scale. These quantities allow the construction of a consistent solar-disk mask and ensure that the segmentation is performed only within the physically meaningful solar-disk region.

The output of the data-processing procedure is a binary coronal-hole candidate mask for each AIA image. The detected candidate regions are generated directly from the standardized AIA 193 Å images through the proposed image-processing pipeline. Therefore, the resulting masks should be interpreted as

image-derived low-emission candidate regions rather than externally verified coronal-hole boundaries.

It should be emphasized that no external event-labeling data, manually annotated reference masks, solar-wind measurements, or magnetic-field measurements are used to determine the segmentation result. This design ensures that the method remains self-contained and reproducible using only publicly available AIA image data and their associated metadata.

3. Method

3.1. Overview of the Proposed Pipeline

The proposed image-based extraction pipeline consists of six major steps: intensity rescaling, dynamic solar-disk radius extraction and mask construction, radial background correction, adaptive K-Means thresholding, hybrid morphological filtering, and area-distance based component selection. The overall objective is to obtain a binary candidate mask that preserves coherent low-emission structures on the solar disk while removing off-disk background pixels, bright active-region effects, small isolated artifacts, and fragmented non-target regions.

Let $D(x, y)$ denote the original AIA 193 Å image intensity at pixel coordinate (x, y) . The final output of the proposed method is a binary candidate mask $M_{\text{final}}(x, y)$, where $M_{\text{final}}(x, y) = 1$ indicates a detected coronal-hole candidate pixel and $M_{\text{final}}(x, y) = 0$ indicates a non-candidate pixel. Since no manually annotated reference mask or external coronal-hole catalog is used in this study, the detected regions should be interpreted as image-derived coronal-hole candidate regions rather than externally verified coronal-hole boundaries.

3.2. AIA Image Intensity Rescaling

The AIA 193 Å images exhibit a high dynamic range. If the raw image intensity is displayed or processed using a purely linear scale, low-intensity coronal-hole candidate regions may be compressed by bright active-region structures, making low-emission boundaries less distinguishable. To improve the separability of low-intensity structures before segmentation, this study applies an intensity-rescaling procedure based on the default AIA visualization normalization implemented in SunPy (The SunPy Community et al. 2020).

The raw image intensity D is first normalized according to the display limits $v_{\min}$ and $v_{\max}$:

$$I(x, y) = \frac{D(x, y) - v_{\min}}{v_{\max} - v_{\min}}, \quad (1)$$

where $I(x, y)$ denotes the linearly normalized intensity. The normalized image is then transformed using an inverse hyperbolic sine stretch:

$$S(x, y) = \frac{\text{arcsinh}(I(x, y)/\alpha)}{\text{arcsinh}(1/\alpha)}, \quad (2)$$

where $S(x, y)$ is the transformed intensity and α is the stretch parameter. Following the AIA display normalization used in SunPy, this study sets $\alpha = 0.01$ (The SunPy Community et al. 2020). This transformation enhances low-intensity regions while compressing high-intensity structures, thereby improving the visibility of candidate-region boundaries for subsequent thresholding.

It should be emphasized that this step is not a spatial sharpening filter. Instead, it is an intensity-scale transformation designed to reduce the dominance of high-intensity structures in high-dynamic-range images. The transformation is fully reproducible and is applied directly to the image-intensity matrix before segmentation.

3.3. Dynamic Solar-Disk Radius Extraction and Mask Construction

Before candidate-region segmentation, the solar-disk region is first extracted in order to exclude off-disk background pixels. Since the apparent solar radius and disk-center position may vary slightly among AIA images due to observational geometry, image registration, and header-dependent plate scale, this study does not use a fixed pixel radius for all images. Instead, the solar-disk extraction range is dynamically determined for each image using the corresponding FITS header information.

Let RSUN_OBS denote the observed solar angular radius recorded in the FITS header, and let CDEL1 denote the angular pixel scale along the image x -axis. The solar radius in pixel units for each image is computed as

$$R_{\text{pix}} = \frac{\text{RSUN_OBS}}{\text{CDEL1}}. \quad (3)$$

This radius is therefore image-specific rather than fixed across the dataset. For Level 1.5 AIA images, the reference-pixel coordinates CRPIX1 and CRPIX2 provide the solar-disk center in the image coordinate system. We denote the disk-center coordinates by

$$(x_c, y_c) = (\text{CRPIX1}, \text{CRPIX2}). \quad (4)$$

To avoid excessive truncation of the solar limb caused by rounding errors or minor boundary uncertainty, a small buffer b is added to the estimated radius. The final extraction radius is defined as

$$R_{\text{final}} = R_{\text{pix}} + b, \quad (5)$$

where $b = 1$ pixel in the present implementation. The dynamic solar-disk mask $M_{\odot}(x, y)$ is then constructed as

$$M_{\odot}(x, y) = \begin{cases} 1, & (x - x_c)^2 + (y - y_c)^2 \leq R_{\text{final}}^2, \\ 0, & \text{otherwise.} \end{cases} \quad (6)$$

Only pixels satisfying $M_{\odot}(x, y) = 1$ are retained as the valid solar-disk region. Pixels outside this dynamically extracted disk range are excluded from

subsequent radial background correction, adaptive thresholding, and morphological filtering. This procedure ensures that the segmentation is performed only within the physically meaningful solar-disk area and prevents the dark off-disk background from being misclassified as coronal-hole candidate regions.

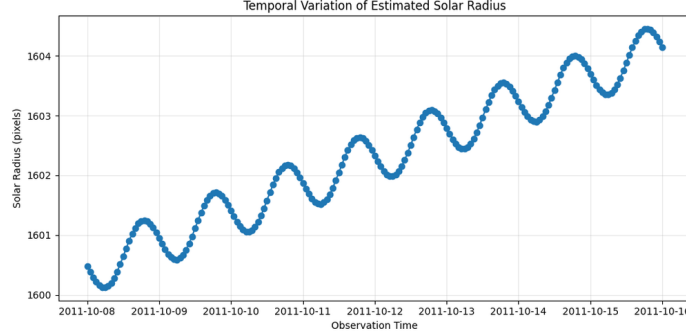


Figure 1. Temporal Variation of Estimated Solar Radius

3.4. Radial Background Intensity Correction

Solar full-disk images often exhibit systematic radial intensity variation across the solar disk. This variation may be caused by limb-related effects, instrumental response, and large-scale brightness gradients (??). If no correction is applied, a single intensity threshold may behave differently near the disk center and near the limb. To reduce this disk-scale variation, a radial background model is estimated and used to normalize the intensity field.

First, pixels inside the solar disk are randomly sampled from the transformed image $S(x, y)$. To reduce the influence of active regions and low-emission candidate regions, only pixels with intensities between the 20th and 80th percentiles are retained as representative quiet-Sun samples. Let $r(x, y)$ denote the normalized radial distance from pixel (x, y) to the solar-disk center:

$$r(x, y) = \frac{\sqrt{(x - x_c)^2 + (y - y_c)^2}}{R_{\text{final}}}. \quad (7)$$

A fourth-order polynomial is then fitted to the selected quiet-Sun samples to estimate the radial background intensity:

$$B(r) = \beta_0 + \beta_1 r + \beta_2 r^2 + \beta_3 r^3 + \beta_4 r^4. \quad (8)$$

The corrected image is defined as

$$I_{\text{corr}}(x, y) = \frac{S(x, y)}{\max\{B(r(x, y)), \epsilon\}}, \quad (9)$$

where ϵ is a small positive constant used to avoid numerical instability. In this study, $\epsilon = 10^{-5}$. The purpose of this correction is to reduce large-scale radial intensity variation while preserving local low-intensity candidate structures.

3.5. Adaptive K-Means Thresholding

After radial correction, an adaptive threshold is computed for each image using K-Means clustering (Lloyd 1982; Reiss et al. 2015). The corrected pixel intensities within the solar disk are grouped into $K = 3$ clusters. This choice is motivated by the typical intensity composition of AIA 193 Å images: low-intensity structures such as coronal holes and filaments, intermediate-intensity quiet-Sun regions, and high-intensity active-region structures.

Let C_0 , C_1 , and C_2 denote the three cluster centers sorted in ascending order:

$$C_0 < C_1 < C_2. \quad (10)$$

Because coronal holes generally appear as low-intensity regions in AIA 193 Å images, the candidate-segmentation threshold is defined using a weighted combination of the darkest and intermediate cluster centers:

$$T_{\text{dyn}} = 0.70C_0 + 0.30C_1. \quad (11)$$

To avoid unstable thresholds under extreme intensity distributions, the threshold is constrained to a relative-intensity interval:

$$T_{\text{clip}} = \min \{ \max (T_{\text{dyn}}, 0.35), 0.65 \}. \quad (12)$$

The initial coronal-hole candidate mask is then defined as

$$M_{\text{ch}}(x, y) = \begin{cases} 1, & I_{\text{corr}}(x, y) < T_{\text{clip}} \text{ and } M_{\odot}(x, y) = 1, \\ 0, & \text{otherwise.} \end{cases} \quad (13)$$

This adaptive thresholding step allows the segmentation criterion to respond to the intensity distribution of each individual image rather than relying on a fixed global threshold.

3.6. Hybrid Morphological Filtering

The initial binary candidate mask may contain fragmented artifacts, irregular boundaries, and isolated small regions. To improve the structural coherence of the candidate mask, a hybrid morphological filtering procedure is applied (Serra 1982).

First, a morphological opening operation with a 5×5 square structuring element is applied to remove isolated noisy pixels and smooth small irregularities:

$$M_{\text{open}} = \text{Open}(M_{\text{ch}}, K_{5 \times 5}). \quad (14)$$

Second, an erosion operation with an 11×11 structuring element is applied to obtain a conservative core-region mask:

$$M_{\text{core}} = \text{Erode}(M_{\text{open}}, K_{11 \times 11}). \quad (15)$$

Connected components in M_{open} are then evaluated according to their overlap with M_{core} . For a connected component A_i , the overlap ratio is defined as

$$R_{\text{overlap}}(A_i) = \frac{|A_i \cap M_{\text{core}}|}{|A_i|}, \quad (16)$$

where $|\cdot|$ denotes the number of pixels in a region. A component is retained if

$$R_{\text{overlap}}(A_i) \geq 0.5. \quad (17)$$

This hybrid strategy differs from directly applying strong erosion or opening alone. It uses the eroded mask only as a conservative reference for structural reliability, while preserving the original extent of components that contain a sufficiently stable core.

3.7. Area- and Distance-Based Component Selection

After morphological filtering, some small fragmented regions may still remain. To reduce the influence of isolated fragmented regions on the final candidate mask, connected components are further filtered using a two-level area and distance criterion.

Let A_i denote the area of the i -th connected component in pixel units. Components with area larger than or equal to 10 000 pixels are defined as major candidate structures and are retained:

$$A_i \geq 10\,000. \quad (18)$$

Components with area smaller than or equal to 5 000 pixels are removed as isolated small structures. Components with area between 5 000 and 10 000 pixels are treated as candidate structures. For each candidate component, the minimum Euclidean boundary distance to any major component is calculated. If the minimum distance is smaller than or equal to 30 pixels, the candidate component is considered geometrically associated with a major candidate structure and is retained:

$$D_{\text{min}}(A_i, A_{\text{major}}) \leq 30. \quad (19)$$

Otherwise, the candidate component is removed. The final candidate mask M_{final} is obtained by combining all retained major and associated candidate components.

Although the filtering thresholds are implemented in pixel units, their corresponding physical scales can be derived from the AIA plate scale recorded in the FITS header. This conversion may be useful when applying the same filtering criteria to images with different pixel scales.

Algorithm 1 Adaptive Extraction of Coronal-Hole Candidate Regions from SDO/AIA 193 Å Images

Require: Raw AIA 193 Å image D , FITS header information

Ensure: Final coronal-hole candidate mask M_{final}

- 1: Compute the solar-disk center (x_c, y_c) from CRPIX1 and CRPIX2
 - 2: Compute the solar radius R_{pix} using RSUN_OBS/CDELTA1
 - 3: Construct the solar-disk mask $M_{\odot}$
 - 4: Apply linear normalization and Asinh intensity stretching to obtain S
 - 5: Estimate the radial background model $B(r)$ using quiet-Sun samples
 - 6: Compute the corrected image $I_{\text{corr}} = S / \max(B, \epsilon)$
 - 7: Perform K-Means clustering with $K = 3$ on disk pixels
 - 8: Sort cluster centers as $C_0 < C_1 < C_2$
 - 9: Compute $T_{\text{dyn}} = 0.70C_0 + 0.30C_1$
 - 10: Clip the threshold to obtain T_{clip}
 - 11: Generate the initial candidate mask $M_{\text{ch}} = (I_{\text{corr}} < T_{\text{clip}}) \cap M_{\odot}$
 - 12: Apply morphological opening with a 5×5 structuring element
 - 13: Compute the core mask using erosion with an 11×11 structuring element
 - 14: Retain connected components with core-overlap ratio at least 0.5
 - 15: Classify connected components by area
 - 16: Retain major components and candidate components adjacent to major components
 - 17: Return the final candidate mask M_{final}
-

3.8. Algorithm Summary

The complete procedure is summarized in Algorithm 1.

3.9. Evaluation Strategy

The proposed method is primarily evaluated in terms of qualitative segmentation behavior, structural coherence, and stability of the extracted coronal-hole candidate masks. Since no manually annotated pixel-level masks are used to train or tune the proposed pipeline, the detected regions should be interpreted as image-derived coronal-hole candidate regions rather than definitive coronal-hole boundaries.

To provide additional quantitative assessments for representative cases, supplementary reference-map comparisons were performed using the red coronal-hole regions indicated in the Solen.info synoptic maps (Alvestad 2026). The Solen.info map was not used as an input to the segmentation algorithm, nor was it used for parameter tuning. It was used only after segmentation as an external visual reference for pixel-wise comparison. Before comparison, the Solen.info map and all detected binary masks were geometrically calibrated to a common solar-disk coordinate system. The comparison was restricted to the solar-disk region.

Let M_{ref} denote the binary reference mask extracted from the red Solen.info coronal-hole regions, and let $M_{\text{det}}^{(m)}$ denote the binary candidate mask obtained

from segmentation method m . Four overlap-based metrics were calculated for each method:

$$\text{Coverage}_{\text{ref}}^{(m)} = \frac{|M_{\text{ref}} \cap M_{\text{det}}^{(m)}|}{|M_{\text{ref}}|}, \quad (20)$$

$$\text{Precision}_{\text{det}}^{(m)} = \frac{|M_{\text{ref}} \cap M_{\text{det}}^{(m)}|}{|M_{\text{det}}^{(m)}|}, \quad (21)$$

$$\text{IoU}^{(m)} = \frac{|M_{\text{ref}} \cap M_{\text{det}}^{(m)}|}{|M_{\text{ref}} \cup M_{\text{det}}^{(m)}|}, \quad (22)$$

and

$$\text{Dice}^{(m)} = \frac{2|M_{\text{ref}} \cap M_{\text{det}}^{(m)}|}{|M_{\text{ref}}| + |M_{\text{det}}^{(m)}|}. \quad (23)$$

Here, $|\cdot|$ denotes the number of pixels in a binary region. The reference coverage measures the proportion of the Solen.info reference region covered by a given detected mask, while the detected precision measures the proportion of the detected mask overlapping the reference region. The IoU and Dice coefficient summarize the overall spatial agreement between each method-specific detected mask and the Solen.info reference mask (Dice 1945). Because the Solen.info map is not a manually drawn pixel-level ground truth mask and may be based on a different observing passband or processing procedure, these metrics are interpreted as reference-map agreement measures rather than definitive supervised segmentation accuracy.

3.10. Baseline Methods

To examine the segmentation behavior of the proposed pipeline, the proposed method was compared with three baseline intensity-based segmentation approaches: Otsu thresholding, gray-level cumulative thresholding, and standard K-Means clustering. These baseline methods were selected because they are commonly used automatic or semi-automatic approaches for extracting low-intensity regions from solar EUV images and can be implemented without manually annotated training masks.

The baseline methods used in this study include:

1. Otsu's automatic thresholding method;
2. gray-level cumulative thresholding;
3. standard K-Means clustering;
4. the full proposed method.

Otsu thresholding determines an automatic threshold from the image histogram by maximizing the between-class variance (Otsu 1979). This method

provides a simple and objective thresholding rule, but its result may be affected by the global intensity distribution of the full-disk image.

Gray-level cumulative thresholding extracts low-intensity pixels according to the cumulative distribution of image intensity (Krista and Gallagher 2009; Rotter et al. 2012). This approach is intuitive for coronal-hole candidate extraction because coronal holes appear as dark regions in AIA 193 Å images. However, the method remains sensitive to the selected cumulative ratio and to changes in the global brightness distribution.

Standard K-Means clustering groups image pixels into different intensity classes and identifies the low-intensity class as the candidate region (Lloyd 1982). Compared with fixed or histogram-based thresholding, K-Means provides an image-adaptive segmentation rule. However, when applied directly without radial background correction and structural post-processing, the resulting mask may still contain fragmented artifacts or isolated dark regions.

The full proposed method combines radial background correction, adaptive K-Means thresholding, hybrid morphological filtering, and area-distance based connected-component selection. Therefore, the comparison focuses on whether the proposed additional preprocessing and post-processing steps can improve structural coherence, reduce fragmented artifacts, and preserve major low-emission candidate regions.

Since no manually annotated reference mask or external coronal-hole catalog is used in this study, the comparison is not interpreted as a supervised segmentation-accuracy benchmark. Instead, the methods are compared in terms of qualitative segmentation behavior, including fragmentation, limb stability, suppression of isolated artifacts, excessive inclusion of non-target dark regions, and preservation of coherent low-emission candidate structures.

3.11. Parameter Settings

Several empirical parameters are used in the proposed pipeline. These parameters were selected based on preliminary visual inspection, segmentation stability, and the need to preserve major low-emission structures while suppressing small fragmented artifacts.

For adaptive K-Means thresholding, the dynamic threshold was defined as

$$T_{\text{dyn}} = 0.70C_0 + 0.30C_1,$$

where C_0 and C_1 denote the darkest and intermediate cluster centers, respectively. This setting gives greater weight to the darkest intensity class, which is consistent with the low-intensity appearance of coronal-hole candidate regions in AIA 193 Å images.

To avoid unstable thresholds under extreme intensity distributions, the threshold was clipped to the interval $[0.35, 0.65]$. This constraint prevents the threshold from becoming excessively low or excessively high when the image contains unusually bright active regions or large dark background variations.

For morphological filtering, a 5×5 square structuring element was used for the opening operation to remove small isolated artifacts while avoiding excessive

erosion of candidate-region boundaries. An 11×11 structuring element was then used for erosion to construct a conservative core mask. The core-overlap threshold was set to 0.5 so that only components with sufficient structural support were retained.

For component-level filtering, connected components with area larger than or equal to 10 000 pixels were regarded as major candidate structures. Components with area smaller than or equal to 5 000 pixels were removed as isolated small fragments. Components with intermediate area were retained only when their minimum boundary distance to a major component was smaller than or equal to 30 pixels. This criterion was used to preserve fragments that were geometrically associated with major low-emission structures while removing isolated artifacts.

These parameter values were fixed throughout the experiments. Therefore, the reported results should be interpreted as the performance of the proposed pipeline under the selected empirical parameter settings. A more systematic sensitivity analysis over a wider parameter range may be conducted in future work.

4. Results

4.1. Qualitative Segmentation Examples

Figure 2 shows an example of the intensity-rescaling procedure. Compared with the raw linear display, the Asinh-transformed image enhances low-intensity candidate regions while suppressing the dominance of bright active regions. This transformation improves the visual separability of low-emission boundaries and provides a more suitable input for the subsequent segmentation steps.

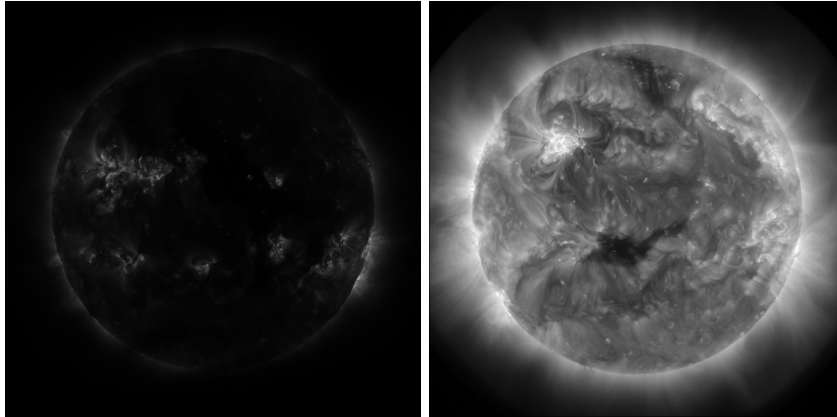


Figure 2. Example of AIA 193 Å intensity rescaling. Left: raw or linearly displayed AIA image. Right: image after Asinh intensity transformation. The nonlinear transformation enhances low-intensity regions while compressing high-intensity active-region structures.

Figure 3 presents an example of the complete candidate-region extraction process. The initial adaptive K-Means mask captures the main low-intensity

regions but may also include fragmented artifacts. After radial correction, hybrid morphological filtering, and area-distance based component selection, the final mask preserves coherent low-emission candidate structures while removing isolated small regions.



Figure 3. Example of the proposed candidate-region extraction pipeline. Left: enhanced AIA 193 Å image. Middle: initial binary candidate mask obtained using adaptive K-Means thresholding. Right: final candidate mask after hybrid morphological filtering and area-distance based component selection.

4.2. Comparison with Baseline Methods

To evaluate the segmentation behavior of the proposed method, three baseline intensity-based segmentation methods were used for comparison: Otsu thresholding, gray-level cumulative thresholding, and standard K-Means clustering. Together with the proposed method, these four results provide a direct comparison between conventional intensity-based segmentation and the proposed structurally refined pipeline. These methods were selected because they are commonly used automatic or semi-automatic segmentation strategies for extracting low-intensity regions from solar EUV images, and they can be implemented without manually annotated training masks.

Since no manually annotated reference mask or external coronal-hole catalog is used in this study, the comparison is not interpreted as a supervised pixel-wise accuracy assessment. Instead, the comparison focuses on qualitative segmentation behavior, including fragmentation, limb stability, suppression of isolated artifacts, excessive inclusion of non-target dark regions, and preservation of major low-emission structures.

Figure 4 shows the full-disk segmentation results obtained using the baseline methods and the proposed method. All segmentation methods were applied to the same AIA 193 Å image so that the differences among the extracted candidate masks can be directly compared.

Otsu thresholding automatically determines the threshold from the image histogram by maximizing the between-class variance. Although this method does not require manual threshold selection, its result may be affected by the global intensity distribution of the full-disk image. In particular, bright active regions, quiet-Sun background, and the relative proportion of dark structures can shift the histogram and influence the selected threshold. As a result, Otsu thresholding may include non-target dark regions or produce fragmented candidate masks.

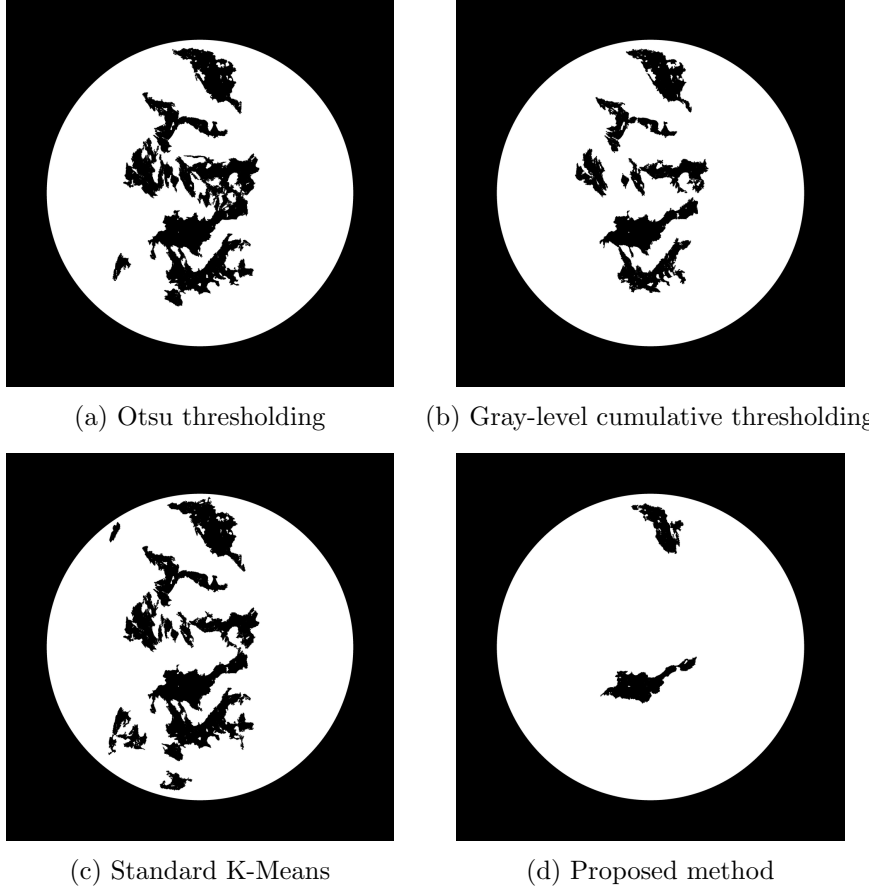


Figure 4. Full-disk comparison of coronal-hole candidate masks obtained using different segmentation methods. Otsu thresholding, gray-level cumulative thresholding, and standard K-Means provide baseline intensity-based segmentation results, whereas the proposed method combines radial background correction, adaptive K-Means thresholding, hybrid morphological filtering, and area-distance based component selection.

The gray-level cumulative thresholding method extracts low-intensity pixels according to the cumulative distribution of image intensity. This approach is intuitive for coronal-hole extraction because coronal holes appear as dark regions in AIA 193 Å images. However, the method still depends on the global intensity distribution. When radial brightness variation or strong active regions are present, the cumulative threshold may not consistently separate coronal-hole candidate regions from other dark or quiet-Sun structures. In addition, this method mainly operates at the pixel-intensity level and does not explicitly consider the spatial coherence of the extracted regions.

Standard K-Means clustering provides an adaptive way to group pixels into different intensity classes. Compared with Otsu thresholding and gray-level cumulative thresholding, K-Means can better reflect the intensity distribution of each individual image. However, when K-Means is applied directly without radial

background correction and structural filtering, the resulting mask may remain sensitive to disk-scale intensity variation. Moreover, standard K-Means only performs intensity-based grouping and therefore may retain fragmented artifacts or isolated dark regions.

In contrast, the proposed method combines adaptive intensity segmentation with additional preprocessing and component-level structural selection. The Asinh intensity transformation enhances low-intensity structures while reducing the dominance of bright active regions. The radial background correction reduces disk-scale intensity variation and improves segmentation consistency between disk-center and limb regions. The hybrid morphological filtering and area-distance based component selection further suppress isolated small regions while preserving major coherent low-emission structures.

Table 1 summarizes the qualitative segmentation behavior of the baseline methods and the proposed method. The comparison indicates that the proposed method provides better structural stability than the baseline intensity-based methods, particularly in terms of fragment suppression and preservation of coherent low-emission candidate regions.

Table 1. Qualitative comparison of segmentation behavior among the baseline methods and the proposed method.

Method	Main advantage	Main limitation	Overall behavior
Otsu thresholding	Automatic threshold selection	Affected by histogram imbalance and global intensity distribution	Moderate stability
Gray-level cumulative thresholding	Direct extraction of low-intensity pixels	Sensitive to the selected cumulative ratio and global intensity distribution	Fragmented regions may remain
Standard K-Means	Image-adaptive intensity clustering	Does not include radial correction or structural post-processing	Good extraction but less stable
Proposed method	Adaptive thresholding with structural filtering	Still primarily image-based and parameter-dependent	Stable structural extraction

Overall, the comparison demonstrates that the proposed method is not simply a direct thresholding procedure. Instead, it integrates intensity enhancement, radial background correction, adaptive K-Means thresholding, and structural post-processing. This combination allows the final mask to retain the main low-emission candidate regions while reducing fragmented artifacts and unstable segmentation behavior. Therefore, the proposed pipeline provides a more stable

and reproducible image-derived mask for coronal-hole candidate extraction from full-disk SDO/AIA 193 Å images.

4.3. Supplementary Reference-Map Evaluation

To complement the qualitative comparison, a supplementary pixel-wise agreement analysis was conducted for the calibrated CH1202 example using the red coronal-hole regions in the Solen.info map as an external reference. The Solen.info reference map and all segmentation results were first transformed to a common solar-disk coordinate system. The red regions in the Solen.info map were extracted as M_{ref} , and the candidate regions obtained from each segmentation method were extracted as method-specific detected masks $M_{\text{det}}^{(m)}$. The evaluation was restricted to the common solar-disk region.

Table 2 summarizes the reference-map agreement metrics for the baseline methods and the proposed method. For each method, the reference coverage indicates how much of the Solen.info reference region was covered by the detected mask, whereas the detected precision indicates how much of the detected region overlapped with the Solen.info reference region. The IoU and Dice coefficient provide overall measures of spatial agreement between each detected mask and the Solen.info reference map.

Table 2. Reference-map agreement metrics between different segmentation methods and the Solen.info reference map for the calibrated CH1202 example.

Method	Reference coverage / Recall	Detected precision	IoU	Dice coefficient
Otsu thresholding	0.989	0.199	0.198	0.331
Gray-level cumulative thresholding	0.979	0.282	0.280	0.437
Standard K-Means	0.960	0.184	0.182	0.308
Proposed method	0.879	0.828	0.744	0.853

The reference-map comparison provides additional quantitative evidence for the spatial correspondence between each segmentation result and the Solen.info indicated coronal-hole regions. Among the baseline methods, standard K-Means achieved the highest reference coverage, indicating that most of the Solen.info reference region was included in the detected mask. However, its detected precision, IoU, and Dice coefficient were substantially lower than those of the proposed method, suggesting substantial over-detection outside the reference regions. Otsu thresholding and gray-level cumulative thresholding also achieved high reference coverage, but their low detected precision indicates that they included many additional non-reference regions.

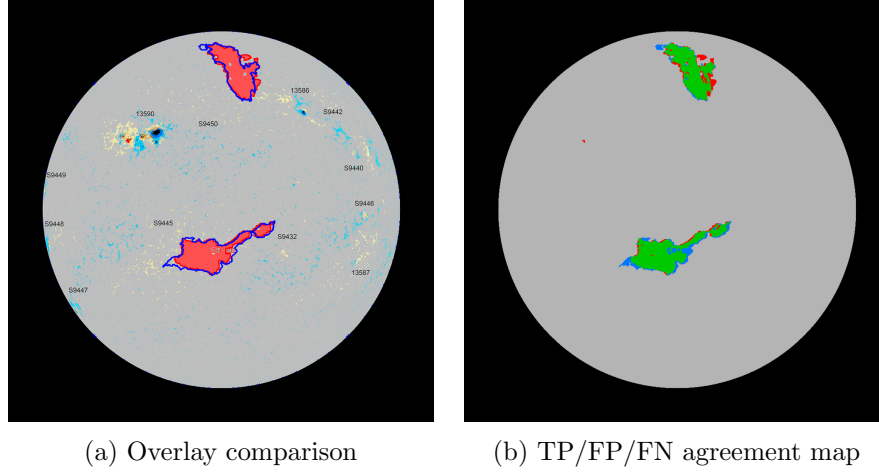


Figure 5. Supplementary reference-map comparison for the calibrated CH1202 example. (a) Overlay comparison between the Solen.info reference regions and the proposed candidate mask. Red shaded regions indicate the coronal-hole regions extracted from the Solen.info reference map, while blue contours indicate the candidate regions detected by the proposed method. (b) Pixel-wise agreement map between the reference and detected masks. Green indicates true-positive overlap, blue indicates false-positive detected regions, red indicates false-negative reference regions, and gray indicates non-candidate background within the solar disk.

In contrast, the proposed method achieved the highest detected precision, IoU, and Dice coefficient. Although its reference coverage was lower than those of the baseline methods, the much higher precision and spatial-overlap scores indicate that the proposed post-processing strategy produced a more spatially concentrated and structurally consistent candidate mask. These results support the qualitative observation that the proposed method reduces excessive inclusion of non-target dark regions while preserving the major coronal-hole-like structures.

To avoid overgeneralizing the comparison from a single calibrated case, an additional CH453 example was also evaluated. This additional case was used to examine whether the baseline methods could also achieve reasonable agreement under more favorable image conditions. Table 3 summarizes the IoU and Dice coefficient for the CH453 and CH1202 examples.

The CH453 example shows that conventional intensity-based methods are not necessarily ineffective. For example, Otsu thresholding achieved an IoU of 0.661 and a Dice coefficient of 0.796, indicating that baseline methods can produce reasonable agreement when the target low-emission region is relatively well separated from the surrounding background. However, the CH1202 example shows a clearer contrast among the methods. In this more challenging case, the baseline methods retained high reference coverage but produced much lower IoU and Dice coefficients, suggesting substantial over-detection outside the reference regions. By contrast, the proposed method maintained higher IoU and Dice values in both examples, supporting its advantage in reducing excessive inclusion of non-target dark regions while preserving the major coronal-hole-like structure.

However, the reported values for both CH1202 and CH453 should not be interpreted as absolute segmentation accuracy because the Solen.info map is not

Table 3. Case-dependent reference-map agreement across two representative examples.

Method	CH453 IoU / Dice	CH1202 IoU / Dice
Otsu thresholding	0.661 / 0.796	0.198 / 0.331
Gray-level cumulative thresholding	0.498 / 0.665	0.280 / 0.437
Standard K-Means	0.765 / 0.864	0.182 / 0.308
Proposed method	0.826 / 0.905	0.744 / 0.853

a manually annotated pixel-level ground-truth mask. In particular, differences in observing wavelength, map-generation procedure, temporal sampling, and boundary definition may lead to discrepancies between the Solen.info regions and the image-derived candidate masks. Therefore, the quantitative metrics are used as supplementary reference-map agreement measures, while the main evaluation remains focused on segmentation behavior, structural coherence, and stability.

4.4. Comparison with Traditional Morphological Opening

To further examine the contribution of the proposed post-processing strategy, the proposed method was compared with a traditional morphological opening operation. This comparison focuses on the post-processing stage rather than on the initial intensity-based segmentation. Therefore, both post-processing strategies were applied to the same initial candidate mask obtained from adaptive K-Means thresholding.

The traditional opening operation is defined as

$$M_{\text{open}} = \text{Open}(M_{\text{ch}}, K_{\text{open}}), \quad (24)$$

where M_{ch} is the initial candidate mask and K_{open} is the structuring element used for the opening operation. In this comparison, the traditional-opening result shown in Figure 6 was generated using a 40×40 square structuring element. This relatively large kernel was used to represent a stronger conventional morphological filtering setting, allowing its effect to be compared with the proposed component-level post-processing strategy. Morphological opening is a common denoising operation for binary images. It can remove small isolated objects and smooth local boundary irregularities. However, when applied directly to fragmented coronal-hole candidate masks, it may also remove thin boundary structures, disconnect weakly connected regions, or fail to distinguish isolated noise from small fragments that are geometrically associated with a major coronal-hole candidate structure.

In contrast, the proposed post-processing method does not rely only on a single opening operation. After the initial opening step, an erosion-based core

mask is constructed to identify structurally reliable regions. Connected components are then evaluated according to their overlap with the core mask. Finally, area-distance based component selection is applied to retain major candidate structures and nearby associated fragments while removing isolated small components.

Figure 6 compares the initial candidate mask, the result after traditional morphological opening, and the final mask obtained using the proposed post-processing strategy. The red boxes mark the local region that is further enlarged in Figure 7. Traditional opening reduces some isolated artifacts, but it may also weaken or remove fragmented boundary structures. In contrast, the proposed post-processing strategy preserves the major low-emission regions while removing isolated components that are not structurally associated with the main candidate regions.

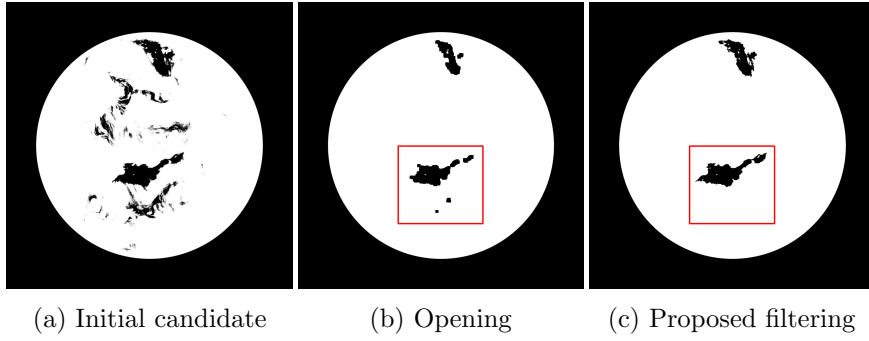


Figure 6. Comparison between traditional morphological opening and the proposed post-processing strategy. The traditional opening operation removes some isolated artifacts but may also erode weakly connected structures. The proposed post-processing method combines core-overlap filtering and area-distance based component selection, leading to a more structurally coherent candidate mask.

To further highlight local structural differences between traditional opening and the proposed post-processing strategy, Figure 7 shows a zoom-in comparison of the same representative region. This comparison focuses on the preservation of small but structurally associated fragments and the removal of isolated artifacts.

The zoom-in comparison indicates that traditional opening can simplify the binary mask, but this simplification may be overly aggressive for fragmented coronal-hole candidate regions. Small structures near the main low-emission region may be removed even when they are spatially associated with the major candidate component. The proposed post-processing strategy is more conservative in this respect because it evaluates connected components using both structural stability and geometric association. Therefore, it can better preserve meaningful fragmented structures while still reducing isolated noise.

This comparison shows that traditional opening alone is insufficient for preserving the structural characteristics of fragmented coronal-hole candidate regions. Although opening is useful as a basic denoising operation, it does not explicitly consider whether a small component is geometrically associated with a major low-emission structure. The proposed method improves upon this lim-

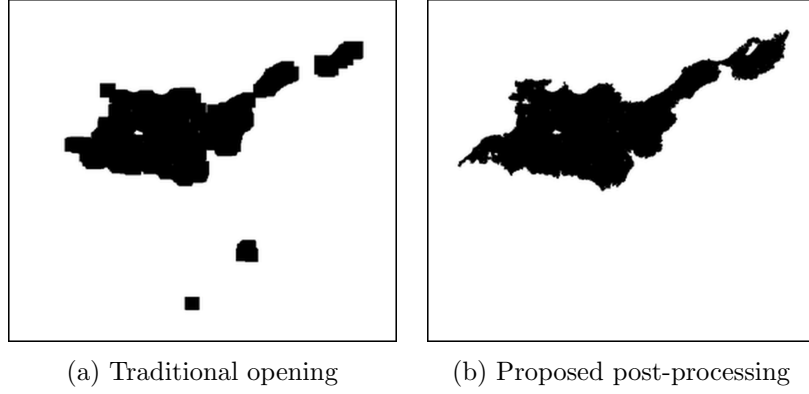


Figure 7. Zoom-in comparison between traditional morphological opening and the proposed post-processing strategy. Traditional opening removes small structures based only on local morphological operations and may therefore erode weakly connected or thin candidate regions. In contrast, the proposed method preserves fragments that are geometrically associated with major low-emission structures while suppressing isolated artifacts.

itation by combining morphological filtering with component-level structural selection.

Table 4. Comparison between traditional morphological opening and the proposed post-processing strategy.

Method	Main effect	Limitation
Traditional opening	Removes small isolated artifacts and smooths local boundary irregularities	May erode weak boundaries or remove fragments associated with major structures
Proposed post-processing	Removes isolated artifacts while preserving geometrically associated structures	Requires empirical area and distance thresholds

4.5. Ablation Analysis

To examine the contribution of each processing step, an ablation analysis was performed by progressively adding radial correction, traditional morphological opening, hybrid morphological filtering, and area-distance based component selection. The purpose of this analysis is to assess how each component changes the segmentation behavior rather than to compute supervised segmentation accuracy.

The ablation results indicate that radial correction improves the consistency of segmentation across the solar disk, especially near the limb. Traditional opening provides basic denoising by removing small isolated artifacts, but it does not

Table 5. Ablation analysis of the proposed coronal-hole candidate extraction pipeline.

Configuration	Main effect	Observed segmentation behavior
Asinh + K-Means	Basic low-intensity extraction	Captures dark regions but may include fragmented artifacts
+ radial correction	Disk-scale intensity normalization	Improves consistency between disk-center and limb regions
+ traditional opening	Basic morphological denoising	Removes small artifacts but may erode weakly connected structures
+ hybrid morphology	Core-based structural filtering	Removes isolated fragments while preserving coherent regions
+ area-distance selection	Component-level selection	Retains major structures and nearby associated fragments

explicitly evaluate whether a fragmented component is structurally related to a major candidate region. Hybrid morphological filtering improves structural reliability by using an erosion-based core mask to identify stable components, while the area-distance selection step further retains candidate fragments only when they are geometrically associated with major candidate components.

5. Discussion

The proposed method provides a reproducible and adaptive pipeline for detecting coronal-hole candidate regions in SDO/AIA 193 Å full-disk images. Compared with fixed-threshold approaches, the method adapts to the intensity distribution of each image through K-Means clustering. Compared with direct clustering alone, the inclusion of radial background correction and hybrid morphological filtering improves the stability and structural reliability of the extracted masks.

A key feature of the proposed method is the combination of pixel-level intensity segmentation and component-level structural selection. Pixel-level thresholding identifies low-emission candidate regions, while component-level filtering removes small isolated artifacts and retains candidate fragments only when they are geometrically associated with major coronal-hole-like structures. This design is useful because coronal holes often exhibit irregular boundaries and fragmented appearances, especially when observed in full-disk EUV images.

The supplementary reference-map evaluation provides an additional quantitative perspective on the spatial agreement between each segmentation result and the Solen.info indicated coronal-hole regions for the calibrated CH1202 example.

The reference coverage reflects the extent to which the proposed method includes the externally indicated regions, whereas the detected precision reflects the extent to which the extracted candidate mask is concentrated within those regions.

The IoU and Dice coefficient further summarize the overall spatial agreement between each method-specific detected mask and the Solen.info reference mask. The additional CH453 comparison provides a more balanced interpretation of the baseline results.

In the CH453 example, some conventional intensity-based methods, especially Otsu thresholding, also achieved reasonable reference-map agreement.

This indicates that baseline methods can perform well when the low-emission target region is relatively distinct from the surrounding background.

However, the calibrated CH1202 example represents a more challenging case, in which the baseline methods tended to include many additional non-reference dark regions.

Under this condition, the proposed method showed a clearer advantage by maintaining substantially higher detected precision, IoU, and Dice coefficient.

Therefore, the benefit of the proposed method should be interpreted not as showing that baseline methods always fail, but as demonstrating improved robustness and structural selectivity in more difficult segmentation conditions.

However, these quantitative metrics should be interpreted with caution. The Solen.info map is used here as an external reference map rather than as a manually annotated pixel-level ground truth. Differences in observing wavelength, map-generation procedure, temporal sampling, spatial resolution, geometric calibration, and boundary definition may contribute to disagreement between the Solen.info regions and the proposed candidate mask. Therefore, the reported coverage, precision, IoU, and Dice values should be interpreted as supplementary reference-map agreement measures, not as definitive supervised segmentation accuracy.

The method also has limitations. First, it relies primarily on EUV intensity and morphology. As a result, dark filament-like structures may occasionally be included in the candidate mask. Incorporating photospheric magnetic-field information, such as SDO/HMI magnetograms, could help distinguish coronal holes from filaments because coronal holes are associated with open magnetic-field regions (Cranmer 2009; ?). Second, several parameters, including the K-Means threshold weight, morphological kernel sizes, overlap-ratio threshold, and area-distance criteria, are empirically selected.

Although sensitivity analysis can support the robustness of the chosen values, further optimization may be required for other instruments, wavelengths, or image resolutions. Third, the method is designed for disk-based segmentation and does not explicitly model three-dimensional coronal geometry. Fourth, although the supplementary Solen.info comparison includes two representative calibrated examples, the quantitative assessment is still limited in scope. Broader evaluation across more events and different solar-disk conditions would be needed to assess the general consistency of the proposed method with external reference maps.

Despite these limitations, the proposed method is useful for generating consistent coronal-hole candidate masks from large sets of SDO/AIA 193 Å images.

Although the present study focuses on candidate-region extraction, the final binary masks may serve as a basis for future structural analysis, including area estimation, centroid tracking, morphology-based characterization, and temporal evolution analysis. In future work, the pipeline may be extended by incorporating magnetic-field constraints, multi-wavelength EUV information, temporal tracking of coronal-hole evolution, and systematic reference-map comparisons across multiple events.

6. Conclusion

This study presents an adaptive image-processing method for coronal-hole candidate extraction in full-disk SDO/AIA 193 Å images. The proposed pipeline integrates intensity rescaling, solar-disk masking, radial background correction, adaptive K-Means thresholding, hybrid morphological filtering, and area-distance based component selection. The objective of the method is to obtain reproducible image-derived candidate masks that preserve coherent low-emission structures while reducing fragmented artifacts and unstable thresholding behavior.

The main conclusions are as follows:

1. The Asinh intensity transformation improves the visibility of low-intensity coronal-hole candidate regions in high-dynamic-range AIA 193 Å images by enhancing dark structures while reducing the dominance of bright active regions.
2. Radial background correction reduces disk-scale intensity variation and improves the consistency of threshold-based segmentation across different radial positions on the solar disk.
3. Adaptive K-Means thresholding provides image-specific segmentation thresholds and avoids the limitations of a fixed global threshold under varying intensity distributions.
4. Hybrid morphological filtering and area-distance based component selection reduce fragmented artifacts while preserving major coherent low-emission candidate structures and nearby associated fragments.
5. The supplementary comparison with Solen.info reference regions provides quantitative overlap measures, including reference coverage, detected precision, intersection-over-union, and Dice coefficient. The additional CH453 example shows that baseline methods can also perform reasonably well under favorable image conditions, whereas the CH1202 example more clearly highlights the advantage of the proposed method in reducing over-detection and improving spatial agreement.
6. The final candidate masks provide a reproducible image-derived representation of low-emission coronal-hole-like structures and may serve as a basis for future structural analysis, temporal tracking, and space-weather-related applications.

Overall, the proposed method provides a reproducible and physically motivated framework for automatic coronal-hole candidate extraction from SDO/AIA

193 Å observations. The Solen.info comparison further supports the use of overlap-based metrics as supplementary reference-map agreement measures. However, because the Solen.info map is not treated as a manually annotated pixel-level ground truth, the reported coverage, precision, IoU, and Dice values should be interpreted as agreement measures rather than definitive supervised segmentation accuracy.

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